Ordinary matter is what we are all made of. It has gravity and interacts with light. Dark matter is a mysterious substance that has gravity but does not interact with light in any known way. Dark energy is a mysterious pressure in the vacuum of space that acts in the opposite direction of gravity, forcing the universe to expand faster than it otherwise would.

What our phrenological exam says is that we understand how the universe behaved, but that most of the universe is made of stuff about which we are clueless. Our profound areas of ignorance notwithstanding, today, as never before, cosmology has an anchor, because the CMB reveals the portal through which we all walked. It's a point where interesting physics happened, and where we learned about the universe before and after its light was set free.

The simple discovery of the cosmic microwave background turned cosmology into something more than mythology. But it was the accurate and detailed map of the cosmic microwave background that turned cosmology into a modern science. Cosmologists have plenty of ego. How could you not when your job is to deduce what brought the universe into existence? Without data, their explanations were just hypotheses. Now, each new observation, each morsel of data, wields a two-edged sword: it enables cosmology to thrive on the kind of foundation that so much of the rest of science enjoys, but it also constrains theories that people thought up when there wasn't enough data to say whether they were right or wrong.

No science achieves maturity without it.

† One nanosecond is a billionth of a second. One picosecond is a trillionth of a second.
†† A. A. Penzias and R. W. Wilson, "A Measurement of Excess Antenna Temperature at 4080 Mc/s," *Astrophysical Journal* 142 (1965): 419–21.